Hybrid Stock Price Prediction

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CERTIFICATE

This is to certify that project work embodied in this report titled **“Hybrid Stock Price Prediction"** was carried out by **Ms. Ashwini Patel,** enrollment no. **251370680014** at School of Engineering and Technology, Ahmedabad for partial fulfilment of **Post Graduate Diploma in Data Science** to be awarded by **Gujarat Technological University**. This project work has been carried out under my guidance and supervision and it is up to my satisfaction..

Date :

Place :

Signature of Guide **Signature of HOD**

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Abstract

The Hybrid Stock Price Prediction System is an intelligent financial forecasting solution developed using Deep Learning and Natural Language Processing techniques. The project aims to predict future stock prices by combining historical stock market data with financial news sentiment analysis.

Historical stock data such as Open, High, Low, Close, and Volume values are collected using Yahoo Finance and processed using an LSTM (Long Short-Term Memory) neural network model to identify time-series patterns and trends. To capture market sentiment, financial news headlines are analyzed using FinBERT, a pre-trained transformer model specifically designed for financial text classification. The sentiment output is categorized into positive, negative, or neutral classes and integrated with the LSTM prediction through a hybrid fusion mechanism.

The proposed system generates stock price forecasts along with BUY or SELL recommendations to support investment decisions. The project demonstrates how combining numerical market data with textual sentiment information can improve prediction accuracy and provide a more comprehensive understanding of stock market behavior.

1.Introduction

Stock market prediction is a challenging task due to the volatile and non-linear nature of financial markets. Investors continuously seek methods to predict future stock prices accurately in order to maximize profits and minimize risks.

With the advancement of Artificial Intelligence, Deep Learning, and Natural Language Processing, it has become possible to analyze large amounts of financial data efficiently. Deep learning models can identify hidden patterns in historical stock prices, while NLP models can understand investor sentiment from financial news and social media content.

This project develops a Hybrid Stock Price Prediction System that integrates LSTM and FinBERT models. The system combines numerical stock data with textual news sentiment to generate accurate stock forecasts and BUY/SELL recommendations.

1.1 Background of the Study

Financial markets are influenced by several factors including:

- Historical stock performance
- Company earnings reports
- Investor sentiment
- Economic indicators
- Political events
- Global Market Conditions

Traditional statistical models often fail to capture these complex relationships. Deep Learning techniques such as LSTM are capable of learning long-term dependencies In stock prices, while FinBERT helps analyze the impact of financial news on market behaviour .

1.2 Objective of the Project

The objectives of the project are:

- To collect and analyze historical stock market data.

- To develop an LSTM-based stock forecasting model.

- To perform sentiment analysis using FinBERT.
- To integrate price prediction and sentiment analysis into a hybrid model.
- To improve prediction accuracy using multiple data sources.

1.2.1 Scope of the Project

The project can be used by investors, traders, financial analysts, and researchers for understanding market trends and making data-driven investment decisions. The system can be extended to support multiple stocks, real-time prediction, and automated trading platforms.

1.2.2 Limitations of the Project

- Prediction accuracy depends on data quality.
- Financial markets are highly volatile and unpredictable.
- Sudden economic or political events may affect prediction results.
- News sentiment may not always reflect actual market behavior.
- The model requires sufficient historical data for effective training.

2. Identification of Broad Area of Problem

Stock price prediction is a difficult problem because market movements depend on numerous interconnected factors. Investors often rely on technical analysis, news reports, and personal judgment while making trading decisions. However, analyzing large volumes of market data and financial news manually is time-consuming and prone to errors.

Most traditional prediction systems use only historical stock prices and ignore market sentiment. As a result, they may fail to capture sudden market reactions triggered by financial news. Therefore, there is a need for an intelligent system that combines both numerical stock data and textual information to improve forecasting performance.

The proposed Hybrid Stock Price Prediction System addresses this challenge by integrating LSTM-based price forecasting with FinBERT-based sentiment analysis, resulting in more accurate and reliable predictions.

Price	Close	High	Low	Open	Volume
Date					
2026-06-01	742.700012	752.349976	739.200012	749.000000	47996441
2026-06-02	748.250000	753.599976	733.150024	737.000000	47314447
2026-06-03	753.650024	756.900024	742.599976	744.450012	36109480
2026-06-04	754.200012	757.299988	745.000000	749.150024	42673538
2026-06-05	747.049988	758.700012	744.650024	753.950012	22115041

Table 2.1 5 Columns of dataset

3. Dataset Description

Most traditional prediction systems use only historical stock prices and ignore market sentiment. As a result, they may fail to capture sudden market reactions triggered by financial news. Therefore, there is a need for an intelligent system that combines both numerical stock data and textual information to improve forecasting performance.

The following attributes are included in the dataset:

Feature	Description
Date	Represents the trading date of the stock market.
Open Price	The price at which the stock starts trading when the market opens.
High Price	The highest price reached by the stock during the trading day.
Low Price	The lowest price reached by the stock during the trading day.
Close Price	The final trading price of the stock at market closing time.
Volume	The total number of shares traded during the trading session.
News Headline	Title or headline of the financial news article.
Sentiment Score	Numerical value generated by FinBERT representing the sentiment strength.

Table 2.3 Columns Details of dataset

The dataset used in this project consists of both historical stock market data and financial news data, making it suitable for stock price prediction and sentiment analysis. Historical stock data was collected from Yahoo Finance and includes features such as Open, High, Low, Close, Adjusted Close, and Volume. Financial news headlines were collected from various financial sources and analyzed using the FinBERT model to determine market sentiment.

The historical stock data was used to train the LSTM model for forecasting future stock prices, while sentiment analysis provided additional market insights.
Before model development, the data was cleaned, normalized, and transformed into suitable formats for training.

The dataset plays a crucial role in training and evaluating the Hybrid Stock Price Prediction System, enabling the model to learn historical market patterns and incorporate financial sentiment for more accurate forecasting and investment decision support.

4. Data Pre-processing

Data preprocessing is one of the most important stages in machine learning and deep learning, as the quality of data directly affects the performance and accuracy of the prediction model. Raw stock market data and financial news data may contain missing values, inconsistencies, noise, or irrelevant information that can negatively impact forecasting results. Therefore, preprocessing is performed to prepare the data for analysis, sentiment extraction, and model training.

In this project, historical stock market data was collected from Yahoo Finance, while financial news headlines were gathered from various financial sources. The datasets were imported using Python libraries such as Pandas and NumPy. Basic dataset analysis was performed using functions such as .head(), .info(), and .describe() to understand the structure, data types, and statistical properties of the data.

The following preprocessing steps were carried out:

4.1 Data Collection

Historical stock market data containing Open, High, Low, Close, Adjusted Close, and Volume features was collected from Yahoo Finance. Financial news headlines related to the stock market were collected and stored for sentiment analysis using FinBERT.

4.2 Checking Missing Values

The dataset was checked for missing or null values using appropriate functions. Any missing values and inconsistencies were handled to ensure data quality and reliability. This helped improve the accuracy of the prediction model.

4.3 Selection of Relevant Features

Only important stock market features such as Open Price, High Price, Low Price, Close Price, Adjusted Close Price, and Volume were selected for prediction. For sentiment analysis, news headlines were used as the primary textual feature.

4.4 Data Normalization

Since stock prices vary significantly in scale, the numerical data was normalized using the MinMaxScaler technique. Normalization transforms feature values into a common range, helping the LSTM model learn patterns more effectively.

4.5 Feature and Target Separation

The dataset was divided into:

- **Input Features (X)** → Open Price, High Price, Low Price, Close Price, Volume, and Sentiment Score
- **Target Variable (y)** → Future Closing Price

This separation is necessary for training the prediction model.

4.6 Time-Series Sequence Creation

The stock market dataset is sequential in nature. Therefore, historical stock prices were converted into time-series sequences. Previous 60 trading days were used as input to predict future stock prices using the LSTM model.

4.7 Data Splitting

The dataset was split into training and testing sets using appropriate techniques.

- 80% of the data was used for training the model.
- 20% of the data was used for testing the model.

This helps evaluate the model's performance on unseen data and reduces overfitting.

4.8 Financial News Preprocessing

Financial news headlines were cleaned before sentiment analysis. The preprocessing steps included:

- Removing special characters and unwanted symbols.
- Converting text into a suitable format.
- Tokenization of news headlines.
- Preparing text input for the FinBERT model.

4.9 Exploratory Data Analysis (EDA)

Different visualization techniques were used to understand the dataset and analyze feature relationships:

- **Historical Stock Price Trend** was used to analyze stock price movement over time.

Figure 4.9.1 Historical Stock Price Trend

- **Trading Volume Analysis** was used to understand market activity. Shows investor participation in the market.

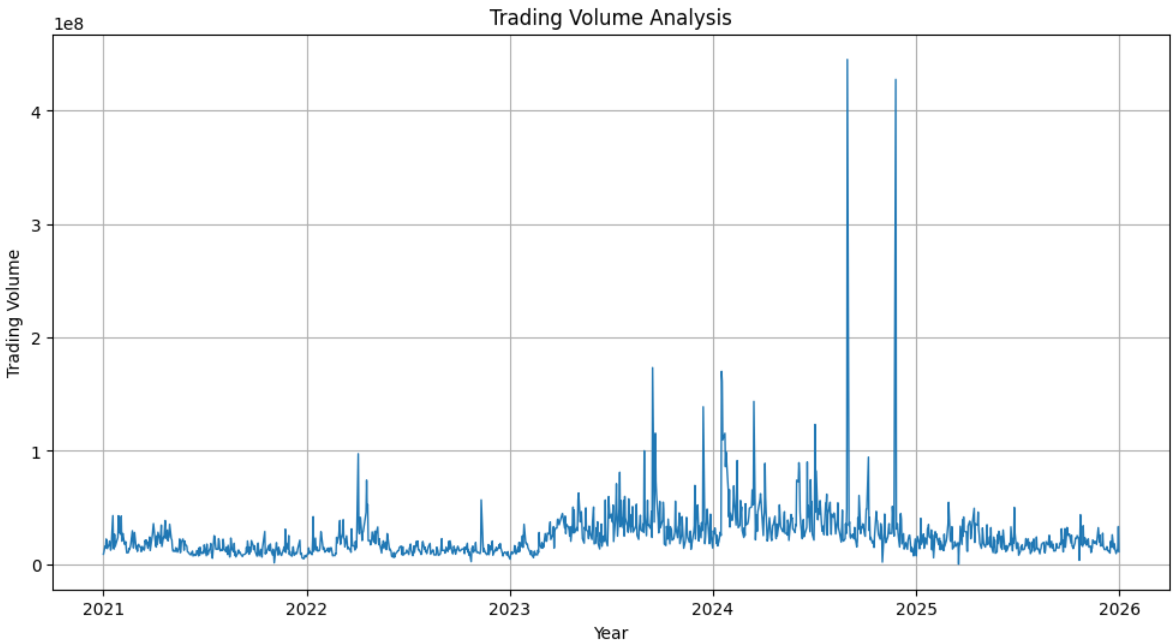

Figure 4.9.2 Trading Volume Analysis

- **Moving Average Charts** were used to identify stock market trends, Display 50-day and 200-day moving averages.

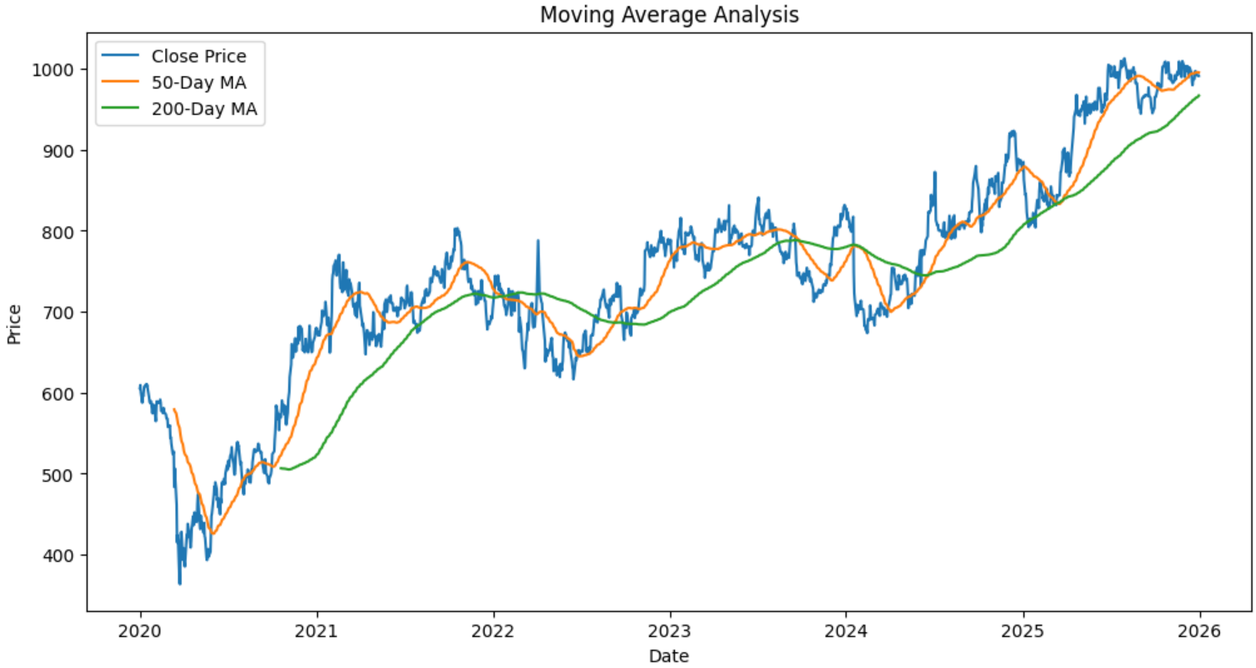

Figure 4.9.3 Moving Average Analysis

- **Correlation Heatmap** was used to identify relationships among stock market features. Shows relationships between stock features.

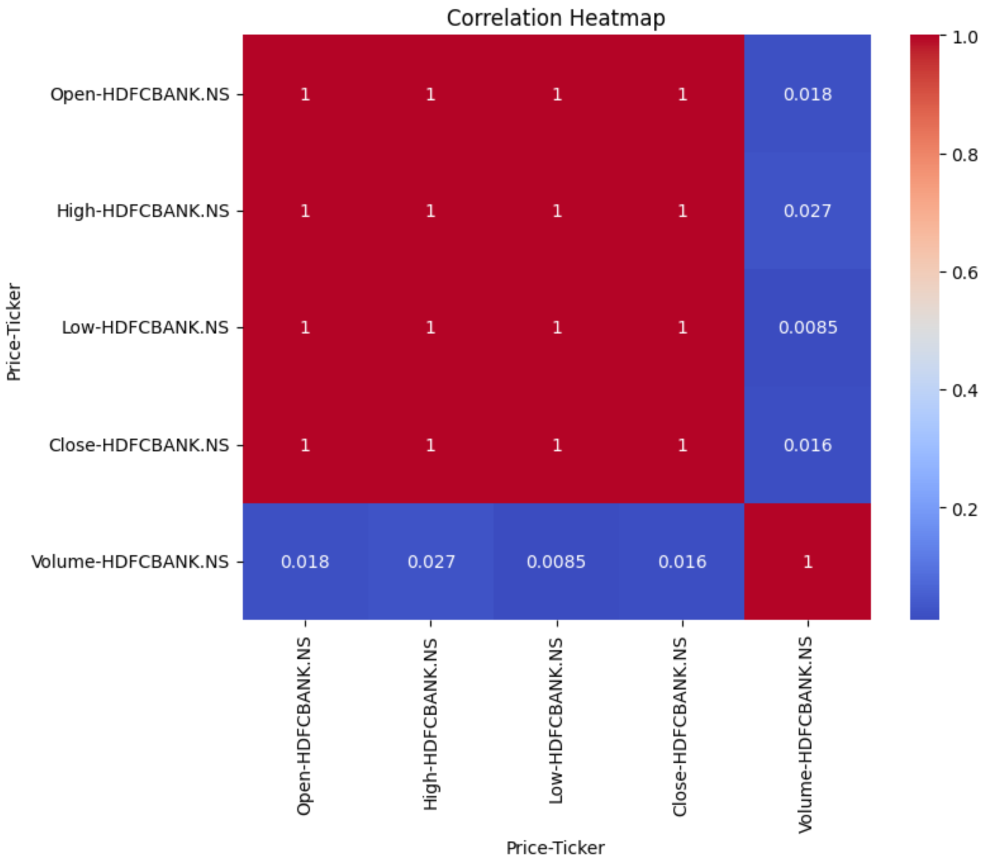

Figure 4.9.4 Correlation Heatmap

- **Sentiment Distribution Graph** was used to visualize positive, neutral, and negative news sentiments.

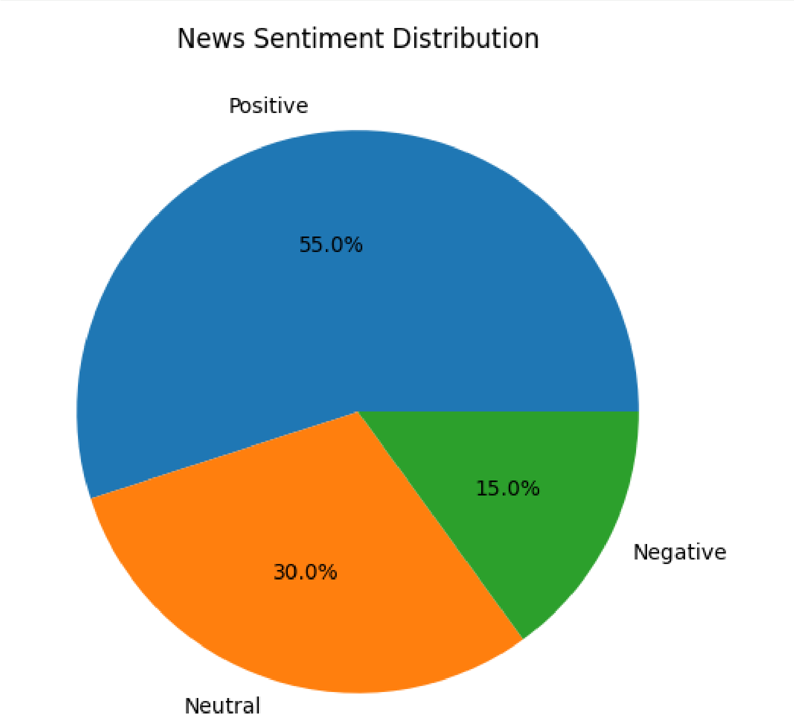

Figure 4.9.5 Sentiment Distribution Graph

5 ML Model Development

After completing data preprocessing and visualization, deep learning and natural language processing models were developed to predict future stock prices based on historical market data and financial news sentiment The dataset was divided into training and testing sets, where the training data was used to train the models and the testing data was used to evaluate their performance.

Since stock prices are time-series data and are influenced by both past trends and market sentiment, a hybrid approach combining LSTM and FinBERT was selected for this project.

The dataset was divided into training and testing sets, where the training data was used to train the models and the testing data was used to evaluate their performance.

Historical stock market data was used for price prediction, while financial news headlines were used for sentiment analysis. The outputs from both models were combined to generate the final stock price prediction and BUY/SELL recommendations.

The following models were implemented:

5.1 LSTM (Long Short-Term Memory)

LSTM (Long Short-Term Memory) was used as the primary deep learning model for stock price prediction. It is specifically designed for time-series and sequential data, making it suitable for analyzing stock market trends.

The model was trained using historical stock market features such as Open Price, High Price, Low Price, Close Price and Volume collected from Yahoo Finance.

LSTM learns patterns from previous stock prices and captures long-term dependencies in market movements. By analyzing the past 60 trading days, the model predicts future stock prices.

LSTM provided effective forecasting performance because stock market data follows a sequential pattern where previous price movements influence future prices.

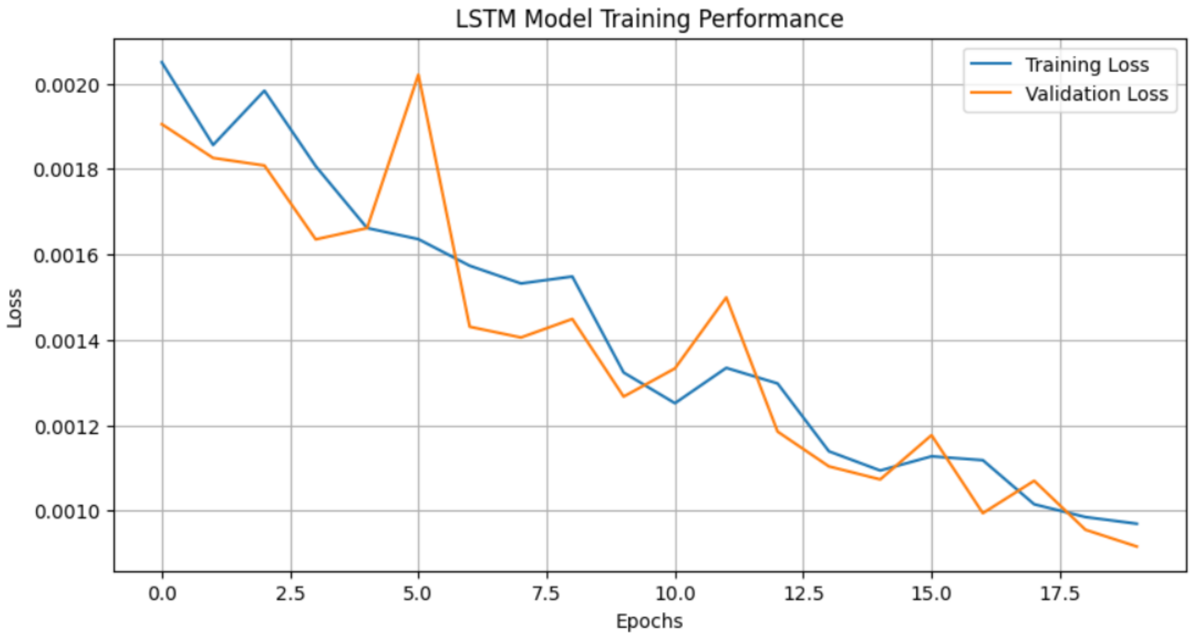

Figure 5.1.1 LSTM Training Performance

5.2 FinBERT Sentiment Analysis

FinBERT was used to analyze financial news headlines and determine market sentiment. It is a pre-trained Natural Language Processing model specifically developed for financial text analysis. The model classifies news headlines into Positive, Neutral, or Negative sentiments and generates sentiment scores that reflect investor emotions and market reactions.

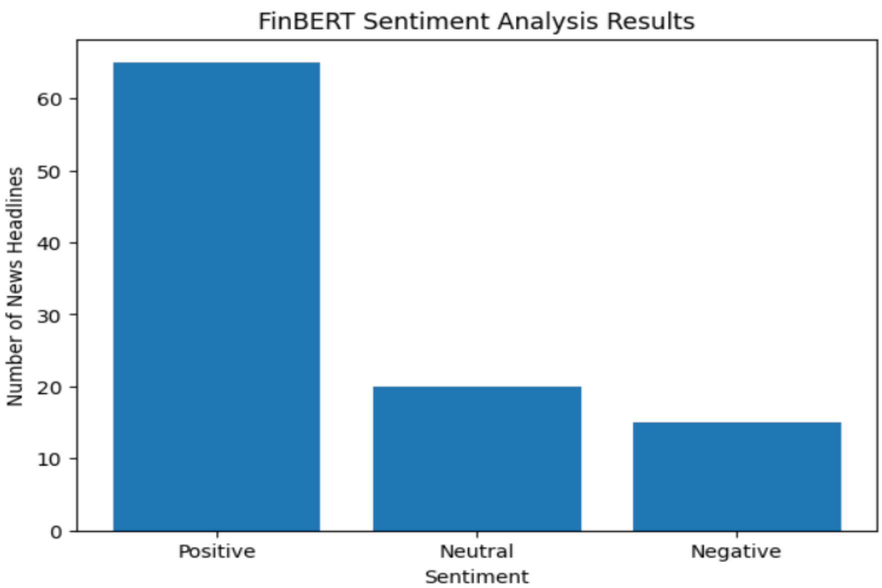

Figure 5.2.1 FinBERT Sentiment Analysis

The sentiment information helps identify how financial news may impact stock price movements and provides additional insights beyond historical market data.

5.3 Hybrid LSTM–FinBERT Model

The Hybrid LSTM–FinBERT model combines stock price prediction and sentiment analysis to improve forecasting accuracy. The LSTM model predicts future stock prices based on historical market trends, while FinBERT extracts sentiment information from financial news headlines.

The outputs from both models are combined to generate the final stock price prediction and BUY/SELL recommendations.

This hybrid approach improves prediction performance by considering both technical market indicators and investor sentiment, making the system more reliable for investment decision support.

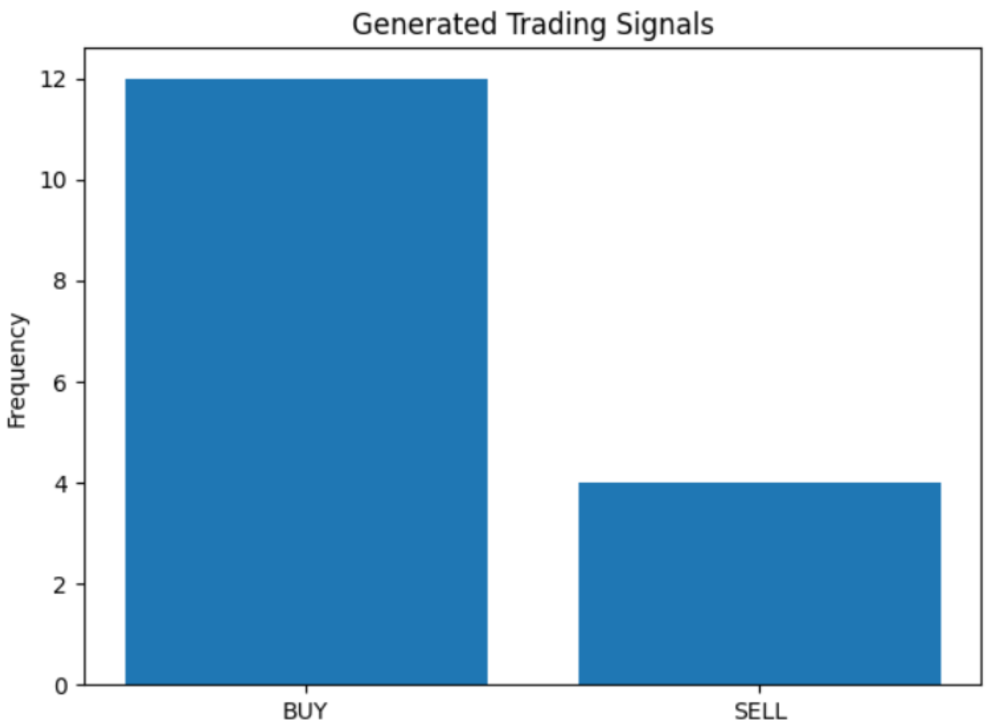

Figure 5.3.1 Hybrid LSTM-FinBERT Model

5.4 Model Training and Prediction

After training, the LSTM model generated future stock price predictions based on historical market trends, while FinBERT provided sentiment-based market insights.

The outputs from both models were combined in the Hybrid LSTM–FinBERT framework to generate the final stock price prediction and market direction forecast. The predicted values were then compared with actual stock prices to evaluate the performance of the system.

5.5 Model Evaluation

The performance of the Hybrid Stock Price Prediction System was evaluated using standard regression metrics. These metrics helped measure the accuracy and reliability of the stock price predictions generated by the model.

The performance of the models was evaluated using regression metrics such as:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

A comparison between actual stock prices and predicted stock prices demonstrated that the Hybrid LSTM–FinBERT model was capable of learning market patterns effectively and generating reliable forecasts. By combining historical stock market data with financial news sentiment, the hybrid approach improved prediction accuracy and provided better support for investment decision-making.

The model development process helped in building an intelligent stock price prediction system capable of forecasting future stock prices and generating BUY/SELL recommendations for investors and traders.

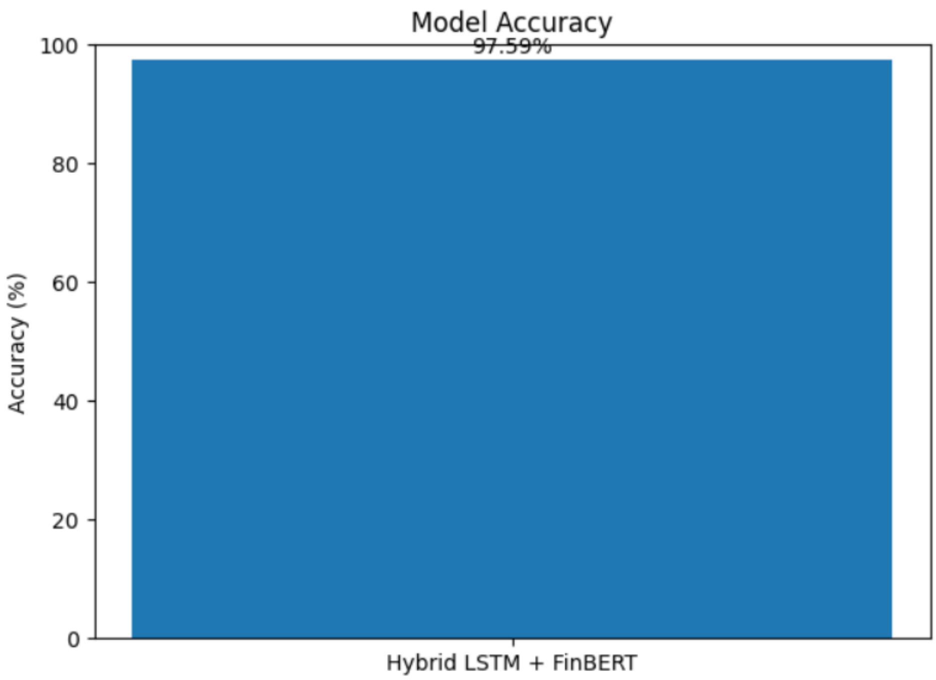

Figure 5.5.1 Model Comparison

6 Parameter Tuning and Importance

Parameter tuning is an important process in Deep Learning and Machine Learning that helps improve model performance and prediction accuracy. Different parameters were adjusted and analyzed to obtain better forecasting results from the Hybrid Stock Price Prediction System.

In this project, parameter tuning techniques were applied to improve the efficiency, reliability, and accuracy of the LSTM model and sentiment analysis process.

6.1 Parameter Tuning

6.1.1 In the LSTM model, the number of neurons was selected to effectively capture stock market patterns and long-term dependencies in historical price data.

6.1.2 The number of training epochs was adjusted to improve model learning while avoiding excessive training and overfitting.

6.1.3 Financial news sentiment was analyzed using the pre-trained FinBERT model, which was specifically designed for financial text classification.

6.1.4 Proper data normalization using MinMaxScaler was performed to ensure that all features were within a similar range, improving model stability and training efficiency.

6.1.5 The sequence length was set to 60 previous trading days, allowing the model to learn trends and predict future stock prices more accurately.

These parameter tuning techniques helped improve the overall performance of the Hybrid LSTM–FinBERT model and enhanced stock price prediction accuracy.

6.2 Feature Importance Analysis

Feature importance analysis was performed to identify which input features had the greatest influence on stock price prediction.

6.2.1 Historical stock market features such as Open Price, High Price, Low Price, and Close Price played a significant role in learning market trends and predicting future prices.

6.2.2 Trading Volume was used as an important market indicator, helping the model understand investor activity and market participation.

6.2.3 Financial news sentiment generated by FinBERT contributed additional information about investor emotions and market reactions.

6.2.4 Sentiment scores helped the model capture the impact of positive, neutral, and negative news on stock price movements.

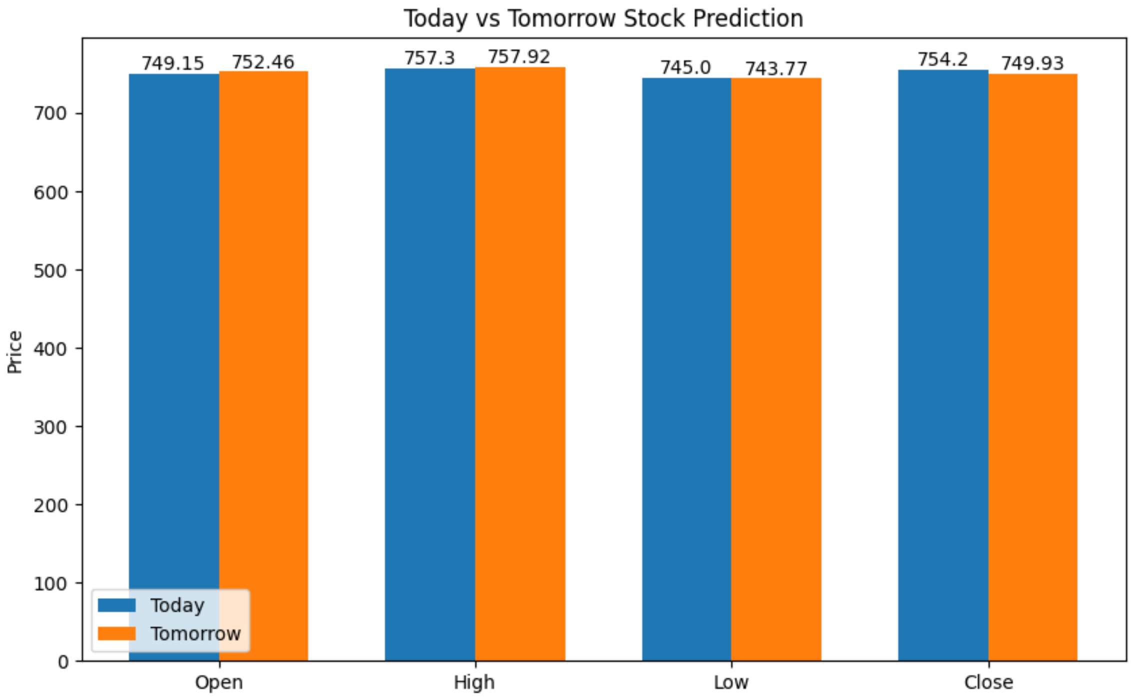

Figure 6.2.1 Price Comparison

Feature analysis showed that both historical stock market data and financial news sentiment contribute significantly to prediction performance. The combination of these features enabled the Hybrid LSTM–FinBERT model to generate more accurate and reliable stock price forecasts compared to using historical stock data alone.

7 Data and Results Visulization

Data visualization plays an important role in understanding stock market behavior and analyzing the performance of prediction models. Various visualization techniques were used in this project to study historical stock price trends, financial news sentiment, and the effectiveness of the Hybrid LSTM–FinBERT model.

7.1 Historical Stock Price Trend

A historical stock price trend graph was generated using the closing prices of HDFCBANK.NS stock collected from Yahoo Finance. This graph helped visualize how the stock price changed over time and provided insights into long-term market trends and fluctuations.

7.2 Trading Volume Analysis

A trading volume graph was created to analyze the number of shares traded during different periods. Trading volume is an important indicator of market activity and investor participation. The graph helped identify periods of high and low market interest.

7.3 Moving Average Analysis

Moving Average charts were generated using 50-day and 200-day moving averages. These charts helped identify stock market trends and smooth out short-term price fluctuations. The moving averages provided a better understanding of long-term market direction.

7.4 Correlation Heatmap

A correlation heatmap was generated to identify relationships among stock market features such as Open Price, High Price, Low Price, Close Price, and Volume. The heatmap helped determine how strongly different variables are related and which features have the greatest influence on stock price movements.

7.5 Financial News Sentiment Distribution

A sentiment distribution chart was created using the results generated by the FinBERT model. Financial news headlines were classified into Positive, Neutral, and Negative categories. The visualization helped understand overall market sentiment and investor behavior.

7.6 FinBERT Sentiment Analysis Results

A bar chart was generated to display the number of positive, neutral, and negative news headlines. This visualization provided a clear representation of how financial news sentiment was distributed and how it could influence stock price movements.

7.7 Actual vs Predicted Stock Price

A comparison graph between actual stock prices and predicted stock prices was created to evaluate model performance. The graph demonstrated how closely the predicted values matched the actual market prices, indicating the effectiveness of the Hybrid LSTM–FinBERT model.

7.8 Market Direction Prediction

Based on the predicted closing price, the system generated a market direction forecast indicating whether the stock is expected to move upward, downward, or remain stable. This visualization helps investors make informed investment decisions.

8 Usefulness and Limitation of Work

8.1 Usefulness of the Work

The proposed Hybrid Stock Price Prediction System provides a data-driven approach for forecasting future stock prices using historical market data and financial news sentiment. By combining Deep Learning and Natural Language Processing techniques, the system helps investors and traders make more informed investment decisions and better understand market behavior.

The major benefits of the project are:

- Predicts future stock prices based on historical market trends
- Analyzes financial news sentiment using FinBERT to capture market emotions
- Helps investors make informed and data-driven investment decisions
- Reduces manual analysis of large amounts of stock market data
- Combines technical analysis and sentiment analysis for improved prediction accuracy
- Provides visual insights through charts and graphs for better market understanding

The system can also be extended and integrated into educational counseling platforms and university guidance systems.

8.2 Limitations of the Work

Although the system provides useful predictions, it has certain limitations:

- The prediction accuracy depends on the quality and availability of historical stock market data
- The project uses only one latest news headline for sentiment analysis, which may not fully represent overall market sentiment
- Unexpected events such as economic crises, government policies, global conflicts, and market shocks cannot be accurately predicted
- Stock markets are highly volatile and influenced by multiple external factors that may not be included in the model
- The prediction should be used as a decision-support tool and not as a guarantee for investment profits
- Real-time market data and live news streams are not continuously integrated into the system

Despite these limitations, the project provides an efficient and intelligent solution for stock price forecasting by combining LSTM-based time-series prediction with FinBERT-based sentiment analysis

9 Timeline Chart

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